Lesson 2.10 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Mean or Median? · I can choose the mean or the median to describe a data set based on its shape.

S2: Matching Center and Spread · I can match a measure of center and a measure of spread to the shape of a data set.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Mean or Median?

I can choose the mean or the median to describe a data set based on its shape. · 6.SP.B.5d

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. mean (no outlier) b. median (65 is an outlier)	Ask whether the decision needs a central observation or an equal share of the total, then compare both summaries.
2	(20 + 22) ÷ 2 = 21 min (the mean, 50 min, is pulled up by 200)	Ask whether the decision needs a central observation or an equal share of the total, then compare both summaries.
3	B — mean	Ask the learner to show why B — mean is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Selects a measure mechanically without connecting it to the question..

Response: Ask whether the decision needs a central observation or an equal share of the total, then compare both summaries.

Also check: Choosing the mean for data with an outlier because it “uses every value.”

Also check: Finding the median without first putting the values in order.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

For 1,2,3,4,20, find mean and median and choose a center to describe a central individual value.

Teacher solution: Mean 6, median 3; median is a defensible choice for a central individual value because the high value raises the mean.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Total 40, mean 8, median 4. Median 4 is a useful central-family description because the large collection raises the mean. Mean 8 gives the equal share for five boxes. The purpose matters.

Misconception response anchor: Selects a measure mechanically without connecting it to the question.. Use ask whether the decision needs a central observation or an equal share of the total, then compare both summaries.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: For ___ I choose the ___, because ___. For a different purpose, ___ could be useful.

Session 2 · Matching Center and Spread

I can match a measure of center and a measure of spread to the shape of a data set. · 6.SP.B.5d

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. mean and MAD b. median and IQR	Sketch the data, compute the candidate summaries, and make the choice match the purpose and actual values.
2	median 9; Q1 = 6, Q3 = 12; IQR = 6	Sketch the data, compute the candidate summaries, and make the choice match the purpose and actual values.
3	C — median with MAD	Ask the learner to show why C — median with MAD is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Pairs a center and spread without inspecting shape or assumes a resistant measure never changes..

Response: Sketch the data, compute the candidate summaries, and make the choice match the purpose and actual values.

Also check: Mixing the pairs (median with MAD, or mean with IQR).

Also check: Missing a low outlier (skewed left) because no value is far above the rest.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

For symmetric data 2,4,6,8,10, give mean and MAD and explain the pair.

Teacher solution: Mean 6; distances 4,2,0,2,4; MAD 2.4. The pair describes center and average distance for the symmetric distribution.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: A is symmetric about 8: mean 8 and MAD 2.4 are useful. B has a large high value: median 8 and IQR 16 are defensible; Q1 5, Q3 21. Do not claim the IQR is unaffected merely because it is resistant.

Misconception response anchor: Pairs a center and spread without inspecting shape or assumes a resistant measure never changes.. Use sketch the data, compute the candidate summaries, and make the choice match the purpose and actual values.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The distribution is ___, so I describe its center with ___ and its spread with ___ because ___.